

```
!pip install langchain langchain-core langchain-community pydantic duckduckgo-search langchain_experimental
```

```
Requirement already satisfied: langchain in /usr/local/lib/python3.11/dist-packages (0.3.24)
Requirement already satisfied: langchain-core in /usr/local/lib/python3.11/dist-packages (0.3.55)
Requirement already satisfied: langchain-community in /usr/local/lib/python3.11/dist-packages (0.3.22)
Requirement already satisfied: pydantic in /usr/local/lib/python3.11/dist-packages (2.11.3)
Requirement already satisfied: duckduckgo-search in /usr/local/lib/python3.11/dist-packages (8.0.1)
Collecting langchain_experimental
  Downloading langchain_experimental-0.3.4-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: langchain-text-splitters<1.0.0,>=0.3.8 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (0.3.11)
Requirement already satisfied: langsmith<0.4,>=0.1.17 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (0.3.31)
Requirement already satisfied: SQLAlchemy<3,>=1.4 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (2.0.40)
Requirement already satisfied: requests<3,>=2 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (2.32.3)
Requirement already satisfied: PyYAML<=5.3 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (6.0.2)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (8.1.0)
Requirement already satisfied: jsonpatch<2.0,>=1.33 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (1.33)
Requirement already satisfied: packaging<25,>=23.2 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (24.2)
Requirement already satisfied: typing-extensions<=4.7 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (4.13)
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (4.0.0)
Requirement already satisfied: dataclasses-json<0.7,>=0.5.7 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (0.6.4)
Requirement already satisfied: pydantic-settings<3.0.0,>=2.4.0 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (2.4.0)
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (0.4.0)
Requirement already satisfied: numpy<=1.26.2 in /usr/local/lib/python3.11/dist-packages (from langchain_experimental) (1.26.2)
Requirement already satisfied: annotated-types<=0.6.0 in /usr/local/lib/python3.11/dist-packages (from pydantic) (0.7.0)
Requirement already satisfied: pydantic-core==2.33.1 in /usr/local/lib/python3.11/dist-packages (from pydantic) (2.33.1)
Requirement already satisfied: typing-inspection<=0.4.0 in /usr/local/lib/python3.11/dist-packages (from pydantic) (0.4.0)
Requirement already satisfied: click<=8.1.8 in /usr/local/lib/python3.11/dist-packages (from duckduckgo-search) (8.1.8)
Requirement already satisfied: primp<=0.15.0 in /usr/local/lib/python3.11/dist-packages (from duckduckgo-search) (0.15.0)
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Requirement already satisfied: aiohappyeyeballs<=2.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp<4.0.0,>=3.8.3) (2.3.0)
Requirement already satisfied: aiosignal<=1.1.2 in /usr/local/lib/python3.11/dist-packages (from aiohttp<4.0.0,>=3.8.3) (1.1.2)
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Requirement already satisfied: frozenlist<=1.1.1 in /usr/local/lib/python3.11/dist-packages (from aiohttp<4.0.0,>=3.8.3) (1.1.1)
Requirement already satisfied: multidict<=7.0,>=4.5 in /usr/local/lib/python3.11/dist-packages (from aiohttp<4.0.0,>=3.8.3) (7.0.0)
Requirement already satisfied: propcache<=0.2.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp<4.0.0,>=3.8.3) (0.2.0)
Requirement already satisfied: yarl<=2.0,>=1.17.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp<4.0.0,>=3.8.3) (1.17.0)
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in /usr/local/lib/python3.11/dist-packages (from dataclasses-json) (3.18.0)
Requirement already satisfied: typing-inspect<=1,>=0.4.0 in /usr/local/lib/python3.11/dist-packages (from dataclasses-json) (0.4.0)
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Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.11/dist-packages (from langsmith<0.4,>=0.1.17) (0.23.0)
Requirement already satisfied: orjson<4.0.0,>=3.9.14 in /usr/local/lib/python3.11/dist-packages (from langsmith<0.4,>=0.1.17) (3.9.14)
Requirement already satisfied: requests-toolbelt<2.0.0,>=1.0.0 in /usr/local/lib/python3.11/dist-packages (from langsmith<0.4,>=0.1.17) (1.0.0)
Requirement already satisfied: zstandard<0.24.0,>=0.23.0 in /usr/local/lib/python3.11/dist-packages (from langsmith<0.4,>=0.1.17) (0.23.0)
Requirement already satisfied: python-dotenv<=0.21.0 in /usr/local/lib/python3.11/dist-packages (from pydantic-settings) (0.21.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2) (3.0.0)
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2) (2.0.7)
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Requirement already satisfied: greenlet<=1 in /usr/local/lib/python3.11/dist-packages (from SQLAlchemy<3,>=1.4) (1.1.3)
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Requirement already satisfied: mypy-extensions<=0.3.0 in /usr/local/lib/python3.11/dist-packages (from typing-inspect<1,>=0.4.0) (0.3.0)
Requirement already satisfied: sniffio<=1.1 in /usr/local/lib/python3.11/dist-packages (from anyio) (1.1)
Downloading langchain_experimental-0.3.4-py3-none-any.whl (209 kB)
209.2/209.2 kB 7.6 MB/s eta 0:00:00
Installing collected packages: langchain_experimental
Successfully installed langchain_experimental-0.3.4
```

✧ Built-in Tool - DuckDuckGo Search

```
from langchain_community.tools import DuckDuckGoSearchRun

search_tool = DuckDuckGoSearchRun()

results = search_tool.invoke('top news in india today')

print(results)
```

The Chinese Foreign Ministry spokesperson Guo Jiakun has condemned the terrorist attack in Jammu and Kashmir's Pahalgam and

```
print(search_tool.name)
print(search_tool.description)
print(search_tool.args)
```

```
duckduckgo_search
A wrapper around DuckDuckGo Search. Useful for when you need to answer questions about current events. Input should be a sea
{'query': {'description': 'search query to look up', 'title': 'Query', 'type': 'string'}}
```

✧ Built-in Tool - Shell Tool

```
from langchain_community.tools import ShellTool

shell_tool = ShellTool()

results = shell_tool.invoke('ls')
```

```
print(results)
```

Executing command:

```
ls
sample_data
```

```
/usr/local/lib/python3.11/dist-packages/langchain_community/tools/shell/tool.py:33: UserWarning: The shell tool has no safewarnings.warn(
```

Custom Tools

```
from langchain_core.tools import tool
```

Step 1 - create a function

```
def multiply(a, b):
    """Multiply two numbers"""
    return a*b
```

Step 2 - add type hints

```
def multiply(a: int, b: int) -> int:
    """Multiply two numbers"""
    return a*b
```

Step 3 - add tool decorator

```
@tool
def multiply(a: int, b: int) -> int:
    """Multiply two numbers"""
    return a*b
```

```
result = multiply.invoke({"a":3, "b":5})
```

```
print(result)
```

```
15
```

```
print(multiply.name)
print(multiply.description)
print(multiply.args)
```

```
multiply
Multiply two numbers
{'a': {'title': 'A', 'type': 'integer'}, 'b': {'title': 'B', 'type': 'integer'}}
```

```
print(multiply.args_schema.model_json_schema())
```

```
{'description': 'Multiply two numbers', 'properties': {'a': {'title': 'A', 'type': 'integer'}, 'b': {'title': 'B', 'type': 'integer'}}
```

Method 2 - Using StructuredTool

```
from langchain.tools import StructuredTool
from pydantic import BaseModel, Field
```

```
class MultiplyInput(BaseModel):
    a: int = Field(required=True, description="The first number to add")
    b: int = Field(required=True, description="The second number to add")
```

```
def multiply_func(a: int, b: int) -> int:
    return a * b
```

```
multiply_tool = StructuredTool.from_function(
    func=multiply_func,
    name="multiply",
    description="Multiply two numbers",
    args_schema=MultiplyInput
)
```

```
result = multiply_tool.invoke({'a':3, 'b':3})
```

```
print(result)
print(multiply_tool.name)
print(multiply_tool.description)
print(multiply_tool.args)
```

```
9
multiply
Multiply two numbers
{'a': {'description': 'The first number to add', 'required': True, 'title': 'A', 'type': 'integer'}, 'b': {'description': 'T
```

Method 3 - Using BaseTool Class

```
from langchain.tools import BaseTool
from typing import Type
```

```
# arg schema using pydantic
```

```
class MultiplyInput(BaseModel):
    a: int = Field(required=True, description="The first number to add")
    b: int = Field(required=True, description="The second number to add")
```

```
class MultiplyTool(BaseTool):
    name: str = "multiply"
    description: str = "Multiply two numbers"

    args_schema: Type[BaseModel] = MultiplyInput

    def _run(self, a: int, b: int) -> int:
        return a * b
```

```
multiply_tool = MultiplyTool()
```

```
result = multiply_tool.invoke({'a':3, 'b':3})

print(result)
print(multiply_tool.name)
print(multiply_tool.description)

print(multiply_tool.args)
```

```
9
multiply
Multiply two numbers
{'a': {'description': 'The first number to add', 'required': True, 'title': 'A', 'type': 'integer'}, 'b': {'description': 'T
```

Toolkit

```
from langchain_core.tools import tool

# Custom tools
@tool
def add(a: int, b: int) -> int:
    """Add two numbers"""
    return a + b

@tool
def multiply(a: int, b: int) -> int:
    """Multiply two numbers"""
    return a * b
```

```
class MathToolkit:
    def get_tools(self):
        return [add, multiply]
```

```
toolkit = MathToolkit()
tools = toolkit.get_tools()

for tool in tools:
    print(tool.name, "=>", tool.description)
```

```
add => Add two numbers
multiply => Multiply two numbers
```

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